

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Mechanical Engineering) SEMESTER - 5 Winter 2025 (Regular)

Course :Bachelor of Technology (Mechanical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTMC503 - THEORY OF MACHINES- II

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question) 12

- 1 Crowning of pulley is done
 - a) To make them more sturdy b) To avoid the slipping of belt c) To make pulley look more pleasant in appearance d) To enable pulley rigidly fixed to the shaft
- 2 The common normal at the point of contact between the mating teeth of the gears must pass through the
 - a) Centre of pinion b) Centre of gears c) Pitch point d) None of the options
- 3 Hour and minute hand are connected in a mechanical clock mechanism by means of
 - a) Simple gear train b) Epicyclic gear train c) Reverted gear train d) None of the options
- 4 Train value of gear train is
 - a) Always less than unity b) Always greater than unity c) Equal to reciprocal of speed ratio d) Equal to speed ratio of gear train
- 5 The internal gear of 60 teeth meshes with a pinion of 20 teeth, module being 6 mm. The centre distance is given by
 - a) 120mm b) 180mm c) 300mm d) 240mm
- 6 For power transmission by belt it is assumed that
 - a) The belt material is elastic b) the length of the belt remains unchanged c) both of the options a & b d) None of the options
- 7 Following type of energy is stored in the flywheel
 - a) Potential energy b) Kinetic energy c) Strain energy d) None of the options
- 8 The gyroscopic couple is due to
 - a) Change in the spin axis b) Change in angular momentum c) Both options a & b d) None of the options
- 9 In automobile, if the vehicle makes a left turn, the gyroscopic couple
 - a) Increase the reaction on the outer wheel b) Does not affect the reaction on the outer wheel c) Decrease the reaction on the outer wheel d) None of the options

- 10 In case of damped vibrations, the amplitude of vibration
a) Increases with time b) Decreases with time c) Remains constant d) None of the options.
- 11 When there is a no change in the amplitude for every cycle of vibration then the body is said to be in
a) Undamped Vibration b) Damped vibration c) Resonance d) None of the options
- 12 Which types of gear box uses gears with non-parallel and non-intersecting axes to connect shaft a) Herringbone gear b) Hypoid gear c) Rack and pinion gear d) Spur gear

Q2. Solve the following.

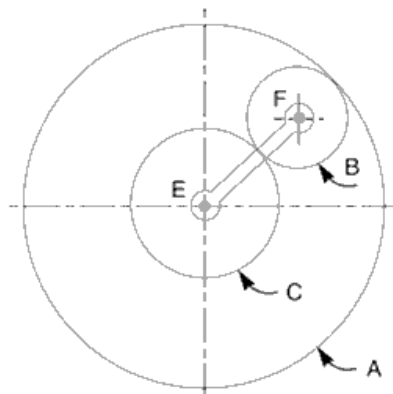
- A) What is centrifugal tension in a belt? Does it affect the power transmission in the belt drives? 6
- B) A belt embraces the smaller pulley over 165° and runs at a speed of 1700 m/min. Dimensions of the belt are width 20 cm, thickness 8 mm. It weighs 1 gm/cm^3 . Determine the maximum power that can be transmitted at the above speed, if the maximum permissible stress in the belt is not to exceed $25 \times 10^5 \text{ N/m}^2$. Take $\mu = 0.25$. 6

Q3. Solve the following.

- A) State & derive law of gearing with neat sketch. 6
- B) A pinion having 20 teeth drives a gear having 56 teeth. The profile of the gears is involute with 20° pressure angle, 4 mm module and Addenda on pinion and gear wheel = 1 module. Find the length of path of contact, arc of contact and the contact ratio. 6

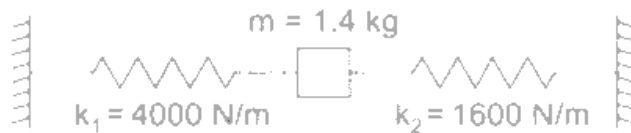
Q4. Solve Any Two of the following.

- A) Explain with neat sketch reverted gear train and compound gear train. 6
- B) Explain the nomenclature of Bevel Gear with neat sketch. 6
- C) An epicyclic gear consists of three gears A, B and C as shown in Fig. The gear A has 72 internal teeth, gear B has 20 external teeth and gear C has 32 external teeth. The gear B meshes with both A and C and is carried on an arm EF which rotates about the center of A at 18 r.p.m in anticlockwise direction. If the gear A is fixed, determine the speed of gear C and its direction of rotation. 6



Q5. Solve Any Two of the following.

- A) The turning moment diagram for a multicylinder engine has been drawn to a scale 1 mm = 600 N-m vertically and 1 mm = 3° horizontally. The intercepted areas between the output torque curve and the mean resistance line, taken in order from one end, are as follows: + 52, - 124, + 92, - 140, + 85, - 72 and + 107 mm², when the engine is running at a speed of 800 r. p. m. If the total fluctuation of speed is not to exceed $\pm 1.5\%$ of the mean, find the necessary mass of the flywheel having radius of gyration of 0.6 m. 6
- B) Explain the turning moment diagram of a four-stroke cycle internal combustion engine. Also explain the role of flywheel in IC Engine. 6
- C) The turbine rotor of a ship has a mass of 8 tonnes and a radius of gyration 0.6 m. It rotates at 1800 r. p. m. clockwise, when looking from the stern. Determine the gyroscopic couple, if the ship travels at 100 km/h and steer to the left in the curve of 75 m radius. 6
- Q6. Solve Any Two of the following.
- A) Explain different types of vibrations with suitable examples. 6
- B) Explain equivalent stiffness for spring in series and parallel combination. Determine the natural frequency of the following spring mass system. 6



- C) A cantilever shaft 60 mm diameter and 200 mm long has a disc of mass 200 kg at its free end. The Young's modulus for the shaft material is 300 GN/m^2 . Determine the frequency of longitudinal and transverse vibrations of the shaft. 6

*** End ***